AIMERS

IBA MBA 68th INTAKE [MATH SECTION 31-55]

[NB: Questions are collected from participating students' memory, IBA DOES NOT publish the question paper. We collected it solely to help future aspirants with their preparation.]

31. Two survey posts, A and B, are located at coordinates (4, -5) and (-8, 7) on a mapping grid. A separate equilateral triangle LMN is constructed so that its height is exactly equal to the distance between posts A and B. What is the area of triangle LMN?

32. In a small town, there were 3300 registered voters. 75 percent of whom voted in an election. If Haroon received 594 votes, what percent of the vote did he receive?

A) 21% B) 24% C) 26% D) 28% E) 32%

33. A researcher records temperature ratios for five different climatic zones across five consecutive days.

Zone A: 14/15, 15/16, 16/17, 17/18, 18/19

Zone B: 15/14, 16/15, 17/16, 18/17, 19/18

Zone C: 5/13, 6/13, 7/13, 8/13, 9/13

In how many of the zones is the arithmetic mean of the five recorded ratios strictly less than the median of those ratios?

A) A only B) B only C) C only D) B and C E) A and B

34. During Fall 2025 semester, 30 percent of the students who enrolled in classes at University of Dhaka enrolled in a science course. If no student enrolled in more than one science course, and if 20 percent of the students who enrolled in science enrolled in chemistry, what percentage of the students who enrolled in classes that semester enrolled in a science course other than chemistry?

A) 6% B) 12% C) 24% D) 14% E) 17%

35. A certain car dealership sells only full-sized and mid-sized cars. One of its sales representatives received an annual salary of Tk 15,000. He also receives a commission of Tk 800 for each full-sized car he sells and Tk 500 for each mid-sized car he sells. What is the least number of cars he must sell in a year to receive total annual earnings of exactly Tk 25,000?

A) 12 B) 12.5 C) 13 D) 14 E) 15

36. An employee of Sicly is entitled to a disability benefit equal to the sum of 60 percent of the first Tk 2000 of monthly salary and 40 percent of the amount of monthly salary in excess of Tk 2000. What is the monthly salary of an employee in Tk who receives Tk 1400 in disability benefits under this plan?

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A) 2,200 B) 2,400 C) 2,240 D) 2,500 E) 3,000

37. An Express travels with an average speed of 100 km/h, taking a 10-minute break every 80 km. Novo travels a break every 2 hours for 2 minutes and travels at an average speed of 50 km/h. If both start at the same time, what distance would Novo Local have travelled by the time Express traveled 600 km?

A) 350 km B) 392 km C) 440 km D) 515 km E) None of these

38. What is the largest integer that can be expressed as a product of four different integers, each of which has a value between -5 and 4 inclusive?

A) 120 B) 240 C) -180 D) -120 E) -60

39. In a video game, a player earns 1 rank point for every 4 monsters defeated, and 1 badge point for every 9 monsters defeated. However, there is no fractional rank point or badge point, implying that defeating 8 monsters earns 2 rank points and zero badge points. If 'n' is a positive integer that represents the number of monsters defeated, how many different values 'n' can have given that the player ends up with the same number of rank points and badge points?

A) 9 B) 10 C) 12 D) 15 E) None of these

40. To unlock a secure laboratory door, researchers must enter a four-digit security code. Each digit can be any number from 0 to 9. The code must satisfy the following conditions:

i. The first digit is 4.
ii. Exactly one of the remaining three digits (positions 2-4) is 1.
iii. All digits must be different (no repetition).

How many valid security codes are possible?

A) 56 B) 112 C) 168 D) 192 E) 216

41. Asif can complete 40% of a project in 10 days, whereas Baset takes 12 days to complete 60% of that project. They work together for 8 days and then Chapol finishes the remaining part in 7 days. How long would it take for three of them to complete 52% of the project if they worked together from the beginning?

A) 2.5 days B) 3 days C) 3.5 days D) 4 days E) 6 days

42. Hamim and Milon were taking turns to shoot air guns. For every 3 shots by Milon, Hamim took 5 shots. The odds of hitting the target by Hamim and Milon are 1/3 and 1/2 respectively. If Milon ends up with 27 missed shots, how many targets did the two hit together?

A) 40 B) 27 C) 50 D) 90 E) 144

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43. A surveyor is mapping three landmarks—P, Q, and R—along a straight rural road, starting from a reference post marked 0, where $0 < P < Q < R$. The gaps between the post and P, between P and Q, and between Q and R follow the proportion 1 : 2 : 3, respectively. Later, she measures that landmark Q is exactly twice as far from the town as landmark P is. Based on this information, what is the absolute coordinate value of landmark R?

A) 3 B) 7 C) 36 D) 45 E) 54

44. A rectangular parking area with sides 40 m and 25 m is surrounded by a walkway of uniform width. The walkway is paved except for a square flowerbed with side 4 m placed at one corner of the walkway. The area of the paved pathway is 566 m^2. What is the width of the walkway in meters?

A) 9 B) 4 C) 5 D) 6 E) None of these

45. Two high-performance servers, Server Alpha and Server Beta, are processing a massive dataset. Working simultaneously at their respective constant speeds, the two servers processed two thirds of the total data in 1 hour and 20 minutes. Suddenly, Server Alpha crashed and stopped working. Server Beta continued working alone at its constant speed and processed the remaining data in 50 minutes. How many hours would it have taken Server Alpha, working alone at its constant speed, to process the entire dataset?

A) 3.5 hrs B) 4 hrs C) 6 hrs D) 10 hrs E) 12 hrs

46. A marketing agency is analyzing the performance of 60 different ad campaigns.

I. 32 campaigns used Social Media Ads.
II. 12 campaigns used neither Social Media nor Email Marketing.
III. Fewer than 35 campaigns used Email Marketing.

Which of the following could be the number of campaigns that used BOTH Social Media Ads and Email Marketing?

A) 17 B) 18 C) 19 D) 14 E) None of these

47. If $y = 2x+3$, which of the following intervals consists of the values of x such that $xy<0$?

A. $0<X<=3/2$ B. $-3/2 <= X<0$ C. $-3/2<X<0$ D. $-3/2<=X<3/2$ E. $-3/2 <X < 3/2$

48. Solution X is five percent salt by volume. to get a solution which is 4% salt, how many cubic c.ms of a solution that is 3.5% salt will have to be added to 500 cubic c.ms of solution X?

A.100 B. 500 C. 950 D. 1000 E. 1250

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49. A set consists of x consecutive integers, including 2. The sum of the integers in the set is -11. What is (are) the possible value(s) of x?

A) 100 B) 500 C) 950 D) A & B E) A, B & C

50. The price of each kindle egg candy is Tk 10. However, a kindle egg candy seller offered customers 1 candy for two returned candy shells. If a boy has Tk 80, how many candies can he get?

A) 8 B) 12 C) 10 D) 15 E) 14

51. If $X = (2-\sqrt{2})/(2+\sqrt{2})$, find $(x-3)^2$

A) 2 B) 6 C) 8 D) 12 E) None of these

52. Five digits — 0, 5, 6, 7, and 9 — are entered into a security panel in some random order, but the system blocks any arrangement that begins with the digit 0. From all the allowed sequences, one is selected at random. What is the probability that the code appearing is 9?

A) 1/5 B) 2/5 C) 1/4 D) 3/16 E) None of these

53. $(25^{7.5} * 5^{2.5})/125^{1.5} = 5^x$ then the value of x is

A) 8 B) 11 C) 13 D) 14 E) 910

54. A factory produced a small batch of 5 hard drives. Unfortunately, due to a machine error, 2 of the drives are defective, while the other 3 are working perfectly. A quality control manager randomly selects 2 hard drives from the batch to test, one at a time (without putting them back). What is the probability that at least one of the tested drives is defective?

A) 1/5 B) 2/5 C)1/2 D) 7/10 E) 9/10

55. When the lines $y = 3x$, $y = -3x + 12$, and $y = 0$ intersect in the coordinate plane, they enclose a triangular region. What is the area of this triangle?

A) 8 B) 10 C) 12 D) 16 E) Cannot be determined

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Written Part- 30 marks, 30 min

1. Aurgumentative writting-15 Marks
"Globalization is not a global friendship, rather global competition and conflict"

2. Explain:-15 Marks
"Need brings Greed, if Greed Spoils it spoils breed"

Courtesy: Mehedi Hasan Nakkhatra

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